

Mirai Botnet Research

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When Everyday Devices Turned Against Us

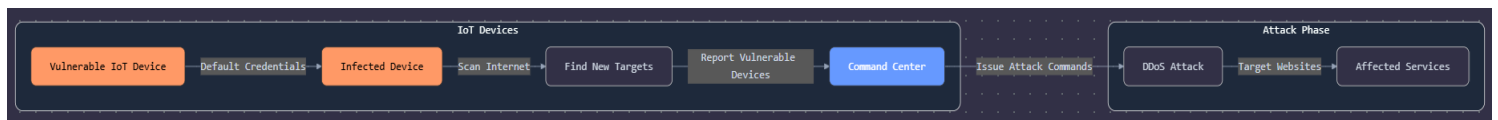
from smart fridges to security cameras look how technology is involved in everything around us from 30 years who could imagine self-driving cars or smart TVs or all the smart devices that have advanced features as automation but let's stop for a moment all of this technologies that help us to do things in easier way and increase every day with newer technologies those things are smart but "smart" doesn't always mean "secure" the truth that those 'iot' devices increase the attack surface that lead for more and sometimes new attacks.

Let's talk about one of those attacks which is Mirai the botnet that transformed iot devices into a global cyber army.

August 2016 a quiet piece of malware suddenly silences the internet this malware was developed to take control of iot devices and take control of them remotely to create armies of 'bots' that flood targets with traffic, causing service outages "distributed denial of service (DDoS) attacks".

Mirai malware was disseminated in source-code format by an alias "Anna-senpai" across an open-forum environment. The code enables remote exploitability of exposed Internet-of-Things (IoT) devices — namely webcams, DVRs, IP cameras, and routers — that can be grouped into botnets and used to mount volumetric DDoS attacks. In early 2017, public sources attributed likely authors of Mirai to be Josiah White and Paras Jha and this how a few lines of leaked code changed cybersecurity forever.

Anatomy of the Beast

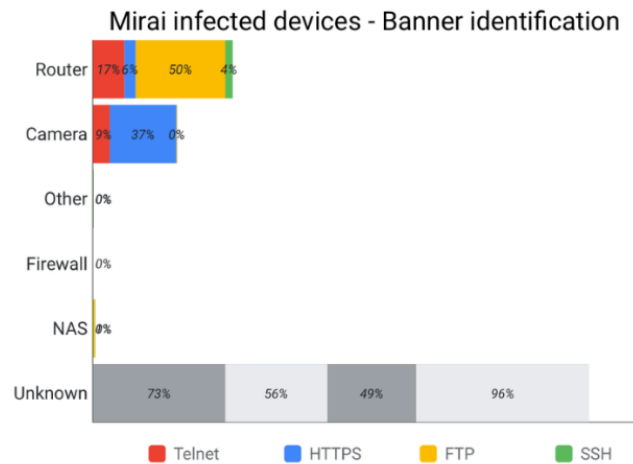


Botnets are groups of Internet devices like computers, cameras, or routers that have been covertly infected with malware. Each infected device turns into a "bot" that the attacker can control remotely. Collectively, those bots can be ordered to do nefarious stuff: send spam, siphon bandwidth, or overwhelm a website until it is knocked offline. Attackers build botnets through the spreading of malware using deceptions like phishing or auto-scanning; once an infected device is within the group, it happens without its owner's awareness.

Mirai looks for internet-connected devices that still use default or weak passwords things like home cameras and routers.

Mirai scans vast regions of IP addresses 'scanning the entire IPv4 address range, looking for devices with open Telnet ports usually port 23 or 2323 or SSH port 22 or HTTP/HTTPS ports 80/443 '. It does so to identify badly secured IoT devices which could be accessed remotely using readily guessable login names and passwords usually default factory usernames and passwords (admin/admin for instance).

NTI Ethical Hacking



When it finds one, the malware installs itself directly in the device's memory, removes any competing malware, so it becomes part of Mirai's army and connects back to central server which is known as a "command & control" server (C&C), the attacker can run commands and tell all those devices to flood a target with traffic, creating a powerful DDoS attack it also supports multiple attack vectors, including UDP floods, TCP SYN floods, and HTTP application-layer attacks. The command-and-control servers then direct the infected devices to flood a target server or service with traffic, overwhelming it and rendering it unavailable.

What makes Mirai especially dangerous is its simplicity. It doesn't rely on advanced hacking techniques or unknown system flaws. Instead, it takes advantage of a basic mistake in IoT security. Mirai also carries the ability to infect every device that is connected to the same network, and therefore it is very likely to develop an enormous botnet that can launch harmful attacks.

Timeline of key events

Aug 2016: Mirai appears, scanning for insecure IoT devices.

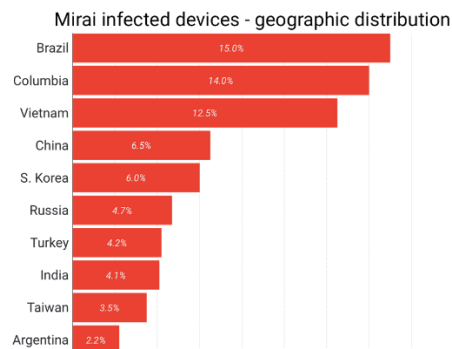
Sep 2016: Launches a 620 Gbps DDoS attack on *KrebsOnSecurity*.

Oct 2016: Attacks Dyn DNS, causing outages across major websites.

Nov 2016: Triggers internet disruptions for UK ISPs like TalkTalk.

Dec 2016: Creators identified, plead guilty; Mirai source code published.

In 2016, this simple idea had a huge impact. Mirai became infamous for launching record-breaking DDoS attacks, starting with a 620 Gbps strike on cybersecurity expert Brian Krebs' website, followed by a 1 Tbps attack on the French company OVH. Later, Mirai hit Dyn a DNS provider knocking popular sites like Twitter, Netflix, Reddit, and PayPal offline across the U.S. and Europe.



Key regions affected by Mirai South America 31.2% of attacking devices, Asia 36.8% of attacking devices (including Russia at 4.7%) and notable presence in Brazil, Vietnam, and Colombia

From Mirai to a Family of Monsters

Since the release of the original Mirai source code, countless variants have emerged each adapting and expanding on the original concept. Today, there are many offshoots of Mirai such as Satori, Okiru, Hakai, Hajime, and Mozi, many of which remain active in attacks. These variants often introduce new capabilities: Satori targets vulnerabilities in Huawei routers, Okiru focuses on ARC processors used in embedded systems, while Hajime and Mozi employ peer-to-peer (P2P) architecture instead of centralized command-and-control (C2) servers. This shift to P2P communication makes them far more resilient to takedown efforts, as there's no single server to disable. Over time, Mirai's descendants have also incorporated more advanced features such as encryption, zero-day exploits, and even ransomware-like components.

The ongoing evolution of these Mirai-based strains underscores the persistent threat posed by insecure IoT ecosystems. If internet-connected devices continue to ship with weak security settings and default credentials, Mirai and its ever-growing family of variants will continue to evolve and remain a powerful weapon in the hands of attackers.

Detection and Defense Fighting an Invisible Enemy

Protecting against Mirai Botnet attacks starts with sound IoT security hygiene. Every single connected device, routers to cameras, must be properly secured and updated. The simplest but most critical step is to change default usernames and passwords during installation. Disabling unused remote-access capabilities such as Telnet and SSH greatly reduces potential entry points, whereas maintaining device firmware current ensures known vulnerabilities are being addressed before the perpetrators can.

Network-wise, businesses have to implement network segmentation for example, isolating IoT devices from infrastructure. This will limit damage if one device is compromised. Ingress and egress filtering need to be implemented in order to prevent infected devices from communicating with the outside world and external command-and-control (C2) servers. At scale, DDoS protection services like those provided by Radware depend on machine learning and behavior-based algorithms to distinguish between legitimate and attack traffic, actively blocking malicious flows without affecting normal user activity.

To detect Mirai infections, behavior-based analysis can detect devices that have anomalous communication behavior or spikes in outbound traffic. Adding threat intelligence feeds and employing honeypots on networks can further help to strengthen early warning systems as well as active research into new variants.

Strengthening defense against Mirai infections ultimately requires good device management and strong network defenses. Below is a quick checklist for risk reduction:

Mirai Defense Checklist:

- Replace all default credentials on IoT devices.
- Disable unnecessary remote services (Telnet, SSH, etc.).
- Regularly update firmware to address vulnerabilities.
- Segregate IoT devices from sensitive networks using VLANs.
- Employ DDoS protection and traffic monitoring utilities.
- Apply ingress and egress filtering to the network perimeter.

While complete immunity from DDoS attacks launched at IoT devices is impossible, orderly security practices can significantly reduce the risk. With an estimated half billion IoT devices — in excess of a quarter billion CCTV cameras worldwide — these core defenses need to become standard practice. Mirai was not the first malware to exploit compromised IoT security, and it won't be the last, but by tightening our defenses, we can make the digital world safer for everyone.

Will Mirai Rise Again?

As the Internet of Things expands at unprecedented levels, so too does its attack surface. Every new connected device, sensor, or smart appliance creates another possible entry point to breach. The next generation of botnets may not just replicate Mirai's brute-force methods they could evolve through AI-driven malware capable of adapting to defenses in real time, choosing optimal attack vectors, and even hiding within normal traffic patterns. To counter this growing threat, nations and industries must collaborate on creating global IoT security standards, promoting "secure-by-design" manufacturing, and investing in continuous threat intelligence and automated defense systems. The struggle against Mirai's successors will depend on the speed at which we can embed security into innovation not as an afterthought.

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